

Over-the-Air Carrier Phase Synchronization for Distributed Arrays: Foundations, the Rashid–Nanzer Consensus Approach, and the Present Project’s Methods and Measurements

Chandler Stevenson

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Abstract

Part I develops the required background from first principles: the coherent-gain arithmetic that motivates phase synchronization, the physics and mathematics of oscillator noise, the estimation theory of phase and frequency measurement from pilots, the identifiability limit of one-way links, and the structure of a closed synchronization loop. Part II gives a complete account of M. Rashid and J. A. Nanzer, “Frequency and Phase Synchronization in Distributed Antenna Arrays Based on Consensus Averaging and Kalman Filtering,” *IEEE Trans. Wireless Commun.* 22(4):2789–2802, 2023: its error model, both algorithms, the derivation logic of its closed-form residual, its reported results, and its assumptions. Part III documents this project’s work: the simulator and its ground-truth grading method, a measured closed-loop error budget with per-term ablation, reciprocal two-way synchronization including the π -ambiguity mechanics, a two-level reimplementation of the paper’s algorithms with a head-to-head comparison that exposes an anti-phase bistability, a two-tier micro-pilot scheme, and a hybrid one-way/two-way calibration scheme validated under Doppler. All quoted numbers are reproducible from `simulation.py` and the test suite in this repository.

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Part I

Foundations

1 Coherent gain and the phase-error requirement

Let N transmitters radiate the same tone at carrier frequency f_c . At a far-field point the fields add as phasors, giving power

$$P \propto \left| \sum_{n=1}^N e^{j\phi_n} \right|^2, \quad (1)$$

where ϕ_n contains each transmitter’s oscillator phase and its propagation phase to the observation point. If all ϕ_n are equal, $P \propto N^2$: N times the power of a single transmitter times the array factor N . For $N = 2$ with phase difference $\Delta\phi$,

$$P \propto |1 + e^{j\Delta\phi}|^2 = 2 + 2 \cos \Delta\phi = 4 \cos^2(\Delta\phi/2), \quad G \equiv \cos^2(\Delta\phi/2), \quad (2)$$

where G is the fraction of ideal two-element gain; $G(0) = 1$, $G(\pi) = 0$. For N elements with independent zero-mean Gaussian phase errors of standard deviation σ , $\mathbb{E}[e^{j\phi_n}] = e^{-\sigma^2/2}$, so the mean coherent power is approximately $N^2 e^{-\sigma^2}$ for large N . Setting $\sigma = 18^\circ = 0.314$ rad gives $e^{-\sigma^2} = 0.906$; this is the origin of the 18° threshold for retaining 90% of ideal gain used throughout the distributed-array literature.

The same arithmetic governs coherent reception. When two stations receive a signal from a passive emitter or scatterer and combine (by summation or cross-correlation), the source’s own phase is common

to both receive chains and cancels; the phase difference between the stations' local oscillators does not. Cross-station coherence is therefore the entire synchronization requirement for passive coherent sensing.

Scale of the problem: at $f_c = 915$ MHz the carrier period is 1.093 ns, so 1 ns of relative timing error corresponds to $360^\circ \times 0.915 = 329^\circ$ of carrier phase. Free-running crystal oscillators disagree by parts in 10^6 (tens of ppm); at 915 MHz, 1 ppm is 915 Hz of carrier offset, accumulating a full cycle of phase error every 1.09 ms. Alignment must be continuously maintained, and for radios connected only by radio, maintained over the air.

2 Oscillator noise

2.1 State model and noise types

Write the oscillator's deviation from nominal as phase ϕ (rad) and angular frequency offset ω (rad/s). Sampled at interval T ,

$$\begin{bmatrix} \phi_{k+1} \\ \omega_{k+1} \end{bmatrix} = \begin{bmatrix} 1 & T \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \phi_k \\ \omega_k \end{bmatrix} + \mathbf{w}_k. \quad (3)$$

The process noise $\mathbf{w}_k$ summarizes several physical mechanisms, classified by the slope of the single-sideband phase-noise spectrum $\mathcal{L}(f) \propto f^{-\alpha}$:

White PM ($\alpha = 0$): each sample of phase carries an independent jitter $\sim \mathcal{N}(0, \sigma_{\text{wpm}}^2)$. An estimator that averages L samples reduces its contribution by $1/L$.

White FM ($\alpha = 2$): the instantaneous frequency is white, so the phase is a random walk. If each sample at rate f_s adds an independent phase step of standard deviation σ_{pn} , then over time t

$$\text{var}(\phi(t) - \phi(0)) = \sigma_{\text{pn}}^2 f_s t \equiv k t. \quad (4)$$

A random walk has independent increments: the walk accumulated between two correction epochs cannot be predicted from anything observed before the first epoch. Consequently kT is an irreducible phase variance for any synchronizer that corrects every T seconds, independent of SNR, estimator, or control law.

Flicker FM ($\alpha = 3$): frequency noise with $1/f$ spectrum. It has no finite-dimensional exact state-space model; the standard surrogate is a sum of first-order Gauss–Markov (AR(1)) processes whose correlation times are log-spaced across the timescales of interest, each contributing a Lorentzian; the sum approximates a $1/f$ spectrum over that range.

Random-walk FM ($\alpha = 4$): white noise added to frequency each step (the σ_ω^2 entry of $\mathbf{w}_k$); models aging and temperature ramps.

2.2 Allan deviation and the numbers used here

The fractional frequency is $y(t) = \dot{\phi}(t)/(2\pi f_c)$. The Allan deviation $\sigma_y(\tau)$ is the standard deviation of the difference of successive τ -averaged values of y , divided by $\sqrt{2}$; each noise type produces a characteristic slope (τ^{-1} for white PM, $\tau^{-1/2}$ for white FM, τ^0 for flicker FM, $\tau^{+1/2}$ for random-walk FM). A quick estimate for white FM: the τ -average of y is $(\phi(t+\tau) - \phi(t))/(2\pi f_c \tau)$, with standard deviation $\sqrt{k\tau}/(2\pi f_c \tau) = \sqrt{k/\tau}/(2\pi f_c)$. The project's default oscillator uses $\sigma_{\text{pn}} = 2 \times 10^{-4}$ rad/sample at $f_s = 1$ MS/s, so $k = 0.04$ rad²/s and $\sigma_y(1\text{ s}) \approx 0.2/(2\pi \cdot 915 \times 10^6) = 3.5 \times 10^{-11}$ — between a good TCXO and an OCXO. The other defaults: white PM 5 mrad/sample; flicker 0.05 Hz RMS (fractional 5.5×10^{-11} floor); frequency random walk 0.1 Hz per 50 ms interval; inter-interval phase process noise 2 mrad.

2.3 One crystal, three errors

A radio synthesizes its carrier and its ADC/DAC sample clock from the same reference crystal. If the reference runs fractionally fast by y , then simultaneously: the carrier is high by $f_c y$ (CFO); the sample clock is fast by y (SFO, in ppm, $10^6 y$); and the receiver's frame timing drifts by $y f_s T$ samples per interval. At the defaults, an initial CFO of 1500 Hz at 915 MHz means $y = 1.64 \times 10^{-6}$: 1.64 ppm of SFO and 0.082 samples of timing drift per 50 ms interval. These are one random process observed three ways and must co-drift in any faithful model; the reference's random walk and flicker move all three together. A digital NCO correction changes none of them — it shifts baseband phase/frequency while the physical crystal keeps drifting — so a model must track the physical (correction-free) frequency separately from the corrected state to derive SFO and timing.

3 Measuring phase and frequency from pilots

3.1 The phase CRLB

Let $r_n = e^{j\phi} s_n + w_n$, $n = 1:L$, with known unit-power pilot s_n and circular complex Gaussian noise of variance σ_w^2 (SNR = $1/\sigma_w^2$). The log-likelihood depends on ϕ through $\frac{2}{\sigma_w^2} \text{Re}\{e^{-j\phi} \sum_n r_n s_n^*\}$; differentiating twice and taking the expectation gives Fisher information $I(\phi) = 2L \text{SNR}$, hence

$$\text{var}(\hat{\phi}) \geq \frac{1}{2 \text{SNR} L}. \quad (5)$$

At 20 dB SNR (SNR = 100) with $L = 2 \times 2047$ pilot samples, the bound is $1.22 \times 10^{-6} \text{ rad}^2$, i.e., 1.1 mrad.

3.2 Frequency from two pilot blocks

Estimate the phase of two identical pilot blocks of L_b samples whose centers are separated by T_{sep} , and form $\hat{\omega} = (\hat{\phi}_2 - \hat{\phi}_1)/T_{\text{sep}}$. Then

$$\text{var}(\hat{\omega}) = \frac{2}{2 \text{SNR} L_b T_{\text{sep}}^2} = \frac{1}{\text{SNR} L_b T_{\text{sep}}^2} \text{ (rad/s)}^2. \quad (6)$$

With $L_b = 2047$, $T_{\text{sep}} = 2175$ samples = 2.175 ms at 1 MS/s, and 20 dB SNR: $\sigma_{\hat{\omega}} = 1.02 \text{ rad/s}$, i.e., 0.16 Hz.

3.3 Phase noise inside the observation

The receiver's LO phase-noise walk runs during the pilot itself. The phase estimate is approximately the average of the walk over the pilot span. For a walk W_i with per-sample variance σ_{pn}^2 averaged over samples $[a, a+n]$, $\text{var}(\bar{W}) \approx \sigma_{\text{pn}}^2 (a + n/3)$ (the first term because the walk enters the window already displaced, the second from averaging a Brownian path). At the default frame (long fields spanning roughly samples 256–4606), $\sigma_{\text{pn}}^2 (256 + 4350/3) \approx 6.8 \times 10^{-5} \text{ rad}^2$: 8.3 mrad, seven times the AWGN bound (5). The same walk decorrelates the two blocks of the frequency estimator, adding variance $\sigma_{\text{pn}}^2 N_{\text{sep}}/T_{\text{sep}}^2 \approx 18 \text{ (rad/s)}^2$: 0.68 Hz, four times the AWGN frequency bound. At these settings estimation error is phase-noise-limited, not SNR-limited, and a filter designed from the oscillator's data sheet must include these terms in its measurement covariance or it will overweight its measurements.

3.4 Waveforms: Zadoff–Chu sequences and the detection metric

A Zadoff–Chu (ZC) sequence of odd length L and root u coprime to L is $x[n] = \exp(-j\pi u n(n+1)/L)$. It has constant amplitude (bounded PAPR through the PA) and ideal periodic autocorrelation (zero at all nonzero cyclic shifts), so a matched filter produces a single sharp timing/phase peak even in multipath, and a cyclic prefix converts linear convolution with the channel into cyclic, preserving the property. The project's frame

is: a short training field (STF), a length-16 ZC repeated 16 times (256 samples); then two long training fields (LTF), each a 128-sample cyclic prefix plus a length-2047 ZC (root 25), total 4606 samples (= 4.6 ms at 1 MS/s).

Detection and coarse CFO use the STF's periodicity. With lag $D = 16$,

$$P(d) = \sum_m r^*[d+m] r[d+m+D], \quad M(d) = \frac{|P(d)|}{\sqrt{\sum |r[d+m]|^2 \sum |r[d+m+D]|^2}}, \quad (7)$$

maximized over d ; M near 1 indicates a repeated structure regardless of channel or CFO. The angle of P estimates CFO: $\hat{\omega} = \angle P / (DT_s)$, unambiguous while $|\omega|DT_s < \pi$, i.e., $|f| < f_s/(2D) = 31.25$ kHz — this sets the acquisition range. Fine timing then maximizes the normalized cross-correlation against the whole known preamble; fine CFO comes from the phase of the correlation between the two LTFs (separation 2175 samples, (6)); the frame phase comes from matched-filtering the LTFs after CFO correction, referenced to the frame center.

3.5 The identifiability limit of one-way measurement

The received pilot phase is

$$m = \underbrace{(\phi_{\text{tx}} - \phi_{\text{rx}})}_{\theta} + \phi_c + n, \quad (8)$$

where ϕ_c is the channel phase (propagation delay modulo the carrier period, plus the multipath composite). θ and ϕ_c enter identically; no processing of one-way data separates them. A one-way synchronizer that drives $m \rightarrow 0$ therefore forces the receiving oscillator to absorb ϕ_c (a constant +2.92 rad at the project's default channel realization). Whether this matters depends on the application:

- If a downstream channel-estimation step exists (a user measuring pilots from both stations, as in cell-free MIMO), any static per-station phase is absorbed into the user's channel estimate; only the *drift* of the differential phase between estimates matters. One-way synchronization suffices.
- With no feedback path (open-loop beamforming at a computed direction, passive coherent sensing), the carriers must align at the antennas, and the ϕ_c bias is disqualifying.

The remedy is reciprocity. Exchanging pilots in both directions over the same channel within a coherence time, with $m_f = \theta + \phi_c + n_1$ and $m_r = -\theta + \phi_c + n_2$,

$$\frac{1}{2}(m_f - m_r) = \theta + \frac{1}{2}(n_1 - n_2), \quad \frac{1}{2}(m_f + m_r) = \phi_c + \frac{1}{2}(n_1 + n_2), \quad (9)$$

and the two combinations are uncorrelated ($\text{cov} = \frac{1}{4}(\text{var } n_1 - \text{var } n_2) = 0$ for equal noise). Each has half the one-way noise variance. Two caveats carry through the rest of this document. First, phases are measured modulo 2π , so the half-difference is defined modulo π : if $(m_f - m_r)$ is computed wrapped, adding 2π before halving shifts the result by π . The branch must be chosen, and a wrong choice is self-consistent (Section 11). Second, only the propagation channel is reciprocal; each node's transmit and receive chains differ. With chain phases $\tau_i^{\text{tx}}, \tau_i^{\text{rx}}$, the forward measurement contains $\tau_A^{\text{tx}} + \tau_B^{\text{rx}}$ and the reverse $\tau_B^{\text{tx}} + \tau_A^{\text{rx}}$, so the half-difference carries a bias $\frac{1}{2}[(\tau_A^{\text{tx}} - \tau_A^{\text{rx}}) - (\tau_B^{\text{tx}} - \tau_B^{\text{rx}})]$ that does not cancel and must come from loopback calibration.

4 The closed loop

4.1 EKF tracking of phase and frequency

The tracked state is the relative offset $x = [\theta, \omega]^\top$ with dynamics (3). To avoid wrap discontinuities the phase is observed through its cosine and sine: the measurement vector is $z = [\cos m, \sin m, \hat{\omega}_{\text{meas}}]^\top$ with model $h(x) = [\cos \theta, \sin \theta, \omega]^\top$ and Jacobian

$$H(x) = \begin{bmatrix} -\sin \theta & 0 \\ \cos \theta & 0 \\ 0 & 1 \end{bmatrix}. \quad (10)$$

Predict: $\hat{x} \leftarrow F\hat{x}$, $P \leftarrow FPF^\top + Q$. Update (iterated, because sin is far from linear at large innovations): starting from the prior $(\hat{x}^-, P^-)$, repeat

$$K = P^- H^\top (HP^- H^\top + R)^{-1}, \quad \hat{x}^{(i+1)} = \hat{x}^- + K[z - h(\hat{x}^{(i)}) - H(\hat{x}^- - \hat{x}^{(i)})] \quad (11)$$

with H re-linearized at $\hat{x}^{(i)}$, until convergence (six iterations suffice here); then the Joseph-form covariance update $P = (I - KH)P^-(I - KH)^\top + KRK^\top$, which preserves symmetry and positive semidefiniteness. Q is assembled from Section 2 (both nodes' random walks, the white-FM walk over one interval, the flicker innovation); R from Section 3 (AWGN CRLB, intra-observation phase noise, white PM).

4.2 Actuation, latency, and the resulting error term

The correction is applied by a numerically controlled oscillator with finite resolution (here: phase quantized to $2\pi/2^{16}$, frequency to 0.01 Hz) and finite processing latency: a command computed from the frame at epoch k loads at epoch $k+\ell$. A controller that knows ℓ commands the *predicted* state $F^\ell \hat{x}$, which cancels the deterministic drift $\omega\ell T$; what cannot be cancelled is (i) the process noise that occurs during the latency (the walk term, already counted per interval) and (ii) the error in the frequency estimate itself, which integrates into phase as

$$\sigma_{\theta, \text{latency}} = \sigma_{\hat{\omega}} \ell T, \quad (12)$$

where $\sigma_{\hat{\omega}}$ is the filter's steady-state frequency posterior. That posterior follows from the scalar Riccati recursion $P^- = P + Q_\omega$, $P = (P^- R_\omega)/(P^- + R_\omega)$: with $Q_\omega = 2(2\pi \cdot 0.1)^2 = 0.79$ and $R_\omega \approx 1.03$ (rad/s)² (the AWGN and phase-noise terms above), the fixed point is $P = 0.59$, $\sigma_{\hat{\omega}} = 0.77$ rad/s, so $\sigma_{\theta, \text{latency}} \approx 0.77 \times 0.05 = 38$ mrad at $\ell = 1$, $T = 50$ ms. Together with the walk term $\sqrt{kT} = 45$ mrad this predicts the measured floors of Part III before any simulation is run.

Part II

The Reference Paper: Rashid & Nanzer, IEEE TWC 2023

5 Setting and error model

The paper treats N nodes forming an open-loop coherent distributed array: nodes coordinate mutually, with no feedback from the beamforming destination. Nodes form an undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. Each node broadcasts its oscillator signal $s_n(t) = e^{j(2\pi f_n t + \theta_n)}$; neighbors estimate (f_n, θ_n) from $L = Tf_s$ received samples. The total phase error at node n (their Eq. 1) is

$$\delta\phi_n = 2\pi \delta f_n T + 2\pi \varepsilon_f T + \theta_n^e + \delta\theta_n^f + \delta\theta_n + \varepsilon_\theta, \quad (13)$$

respectively: phase accrued by the frequency offset over the update interval; by the frequency estimation error; a constant $\theta_n^e \sim \mathcal{U}[0, 2\pi)$ lumping geometry, timing misalignment, and channel-induced phase ("assumed corrected"); intra-interval drift; phase jitter; and phase estimation error. The statistics:

- Frequency drift: $\sigma_f = f_c \sqrt{\beta_1/T + \beta_2 T}$ (white FM plus random-walk FM in Allan-deviation form), with $\beta_1 = \beta_2 = 5 \times 10^{-19}$ for a quartz TCXO.
- Intra-interval drift: $\sigma_\theta^p = \pi T \sigma_f$, from integrating a linearly-varying frequency offset across the interval (their Eqs. 3–4, 13).
- Phase jitter: $\sigma_\theta = \sqrt{2 \cdot 10^{A/10}}$ with integrated phase-noise power $A = -53.46$ dBc, giving 2.7 mrad — *drawn independently every interval*.
- Estimation errors at the CRLBs: $\sigma_f^m = \sqrt{6/((2\pi)^2 L^3 \text{SNR})}$ (normalized frequency; their Eq. 16) and $\sigma_\theta^m = \sqrt{2/(L \text{SNR})}$ (their Eq. 17). Under TDMA with D average neighbors, $L \rightarrow L/D$, inflating both.

6 DFPC: decentralized frequency and phase consensus

Stack node frequencies into $\mathbf{f}$ and phases into $\boldsymbol{\theta}$. One iteration per interval T :

$$\mathbf{f}(k) = \mathbf{W} \mathbf{f}(k-1), \quad \boldsymbol{\theta}(k) = \mathbf{W} \boldsymbol{\theta}(k-1), \quad (14)$$

with the Metropolis–Hastings mixing matrix $w_{ij} = 1/(1 + \max\{n_i, n_j\})$ for $(i, j) \in \mathcal{E}$, $w_{ii} = 1 - \sum_{j \neq i} w_{ij}$, $w_{ij} = 0$ otherwise ($n_i = \text{degree of node } i$). $\mathbf{W}$ is symmetric doubly stochastic; its largest eigenvalue is 1 with eigenvector $\mathbf{1}$, so $\mathbf{W}^I \rightarrow \frac{1}{N} \mathbf{1} \mathbf{1}^\top$ and every node converges to the arithmetic mean of the initial states. Convergence rate and error accumulation are governed by the second-largest eigenvalue modulus λ_2 : sparse graphs have λ_2 near 1 (slow mixing), dense graphs near 0. Row n of (14) involves only node n ’s neighbors, so the computation is local; and because the broadcast *waveform* carries the state, plain DFPC needs no data link — neighbors estimate the state, which is why the estimation errors are injected into the recursion (their Algorithm 1 adds the drift, jitter, and estimation draws each iteration before mixing).

6.1 Derivation of the steady-state residual (their Eq. 27)

With per-iteration error vector $\mathbf{e}_k$ of per-node variance σ_e^2 , the state after I iterations is

$$\mathbf{z}(I) = \mathbf{W}^I \mathbf{z}(0) + \sum_{i=0}^{I-1} \mathbf{W}^{i+1} \mathbf{e}_{I-i}. \quad (15)$$

The first term converges to the average of initial values; the error sum, measured relative to the network average (subtract $\frac{1}{N} \mathbf{1} \mathbf{1}^\top$), has covariance $\sigma_e^2 \sum_{m=1}^I (\mathbf{W} - \frac{1}{N} \mathbf{1} \mathbf{1}^\top)^{2m}$, bounded by $\sigma_e^2 \sum_m \lambda_2^{2m}$. For sparse graphs ($\lambda_2 \rightarrow 1$) the geometric sum is large and the residual saturates at the injected level itself; the total phase spread is then the root-sum-of-squares of the five per-iteration error terms:

$$\sigma_{\phi, \text{total}} = \sqrt{(2\pi\sigma_f T)^2 + (\sigma_\phi^m)^2 + (\pi T \sigma_f)^2 + (\sigma_\theta^m)^2 + \sigma_\theta^2}. \quad (16)$$

Dense graphs mix errors away faster and sit below this bound. Equation (16) is the paper’s central analytical result: a closed-form floor in oscillator data-sheet quantities, CRLBs, and the update interval. Note what is absent: no actuation latency (updates are instantaneous state assignments), no channel term (swallowed by θ_n^e), and the jitter re-drawn per interval has no memory.

7 KF-DFPC: per-node Kalman filtering before averaging

At small T the CRLB terms dominate (16) (fewer samples per estimate). The paper’s remedy: each node runs a two-state Kalman filter on its own noisy state observation, then the network consensus-averages the *posterior means*. State $[f_n, \theta_n]$, process covariance (their Eq. 29)

$$\mathbf{Q} = \begin{bmatrix} \sigma_f^2 & -\pi T \sigma_f^2 \\ -\pi T \sigma_f^2 & \pi^2 T^2 \sigma_f^2 + \sigma_\theta^2 \end{bmatrix}, \quad (17)$$

measurement covariance $\boldsymbol{\Sigma} = \text{diag}((\sigma_f^m)^2, (\sigma_\theta^m)^2)$, standard predict/update (their Eqs. 33, 35–36). After mixing the means with $\mathbf{W}$, the per-node covariances are re-initialized by mixing their entries with squared weights (their Eq. 39). Per-node complexity is $\mathcal{O}(\text{card}\{\chi_n\} + 8)$ per iteration. The variant requires each node to broadcast its posterior mean *and covariance* to neighbors — an explicit data side channel, assumed error-free and instantaneous; plain DFPC does not need one.

Reported results (their Figs. 5–11), at $f_c = 1$ GHz, $f_s = 10$ MHz, $T = 0.1$ ms, SNR = 0 dB, 10^3 trials: KF-DFPC converges in far fewer iterations on sparse graphs (2 vs. 16 at $N=100$, connectivity 0.05; 4 vs. 158 at $N=65$); its converged error is nearly independent of SNR (the filter compensates measurement quality with memory); it outperforms diffusion LMS, diffusion KF, and the Kalman consensus information filter at low SNR and sparse connectivity; and sweeping T shows a DFPC optimum near $T = 20$ ms — drift errors grow with T , estimation errors shrink with T — while KF-DFPC holds the theoretical floor for $T \lesssim 20$ ms. No closed form is derived for KF-DFPC’s steady state; it is characterized by simulation against (16).

8 The paper’s assumptions, stated precisely

(i) The measurement model (their Eq. 15) is $x_n(t) = e^{j(2\pi f_n t + \theta_n)} + n_o(t)$: each node’s oscillator is observed *directly* in AWGN. There is no propagation channel in the measurement, so the identifiability problem of Section 3 cannot arise, and all channel/geometry phase sits in the uniform constant θ_n^e assumed corrected elsewhere. (ii) Consensus updates are instantaneous assignments; no latency, NCO quantization, or control dynamics. (iii) Jitter is i.i.d. per interval; intra- and inter-interval phase noise are statistically disconnected; flicker FM is absent from the analytical model. (iv) Validation is statistics-level Monte Carlo: their Algorithm 1 injects Gaussian draws from the assumed distributions, so simulations test the algebra, not the physics. (v) Time synchronization is assumed complete; the KF side channel is ideal. These are scope choices, not errors; Part III measures their consequences.

Part III

This Project

9 The simulator

The `ota_sync` package simulates the link at sampled-IQ level. Defaults:

carrier	915 MHz	sample rate	1 MS/s
sync interval T	50 ms	SNR (nominal)	20 dB
channel	3GPP TDL-D, 100 ns DS	initial CFO	1500 Hz
shadowing	2 dB, AR(1), 10 s	white FM	2×10^{-4} rad/sample
white PM	5 mrad/sample	flicker FM	0.05 Hz RMS
freq. random walk	0.1 Hz/interval	ADC/DAC	12 bit
NCO quantization	$2\pi/2^{16}$, 0.01 Hz	correction latency	1 interval

Per interval, the transmit chain is: preamble (Section 3) modulated on the master carrier; soft PA limiter; DAC quantization; Sionna TDL channel taps (per-interval snapshots, Jakes-correlated across intervals when mobility is nonzero); AR(1) log-normal shadowing gain; additive noise at a *fixed* thermal floor established once at nominal channel gain, so per-frame SNR fades with the channel (deriving noise from each frame’s received power, a common shortcut, silently pins SNR and removes fading). The receive chain: resampling at the current physical SFO with the accumulated fractional timing offset (whole-sample drift steps the arrival inside the capture window — a receiver re-centering on each detection — and the sub-sample residual is a fractional delay); downconversion by the slave LO; the intra-frame white-FM walk (whose end value is folded back into the oscillator state, plus an independent increment for the dead time, making intra- and inter-interval phase noise one continuous process); white-PM jitter; IQ gain/phase imbalance; AGC; DC offset; ADC quantization. The receiver DSP is the STF/LTF processing of Section 3; the EKF of Section 4 tracks; corrections are forward-predicted by the known latency, quantized, and queued.

9.1 Ground-truth grading: the oracle path

A grading pitfall: an early metric compared the EKF posterior to the measurement that had just updated it. With Kalman gain κ , that residual is $(1 - \kappa)$ times the innovation; at $\kappa \approx 0.9$ it understates the true error by an order of magnitude, and the code reported 0.105 mrad tracking RMSE. The corrected method builds, for every capture, an *oracle twin* sharing everything deterministic — frame, timing jitter draw, channel realization, shadowing state, PA/DAC distortion, sample-clock timing — and omitting everything stochastic or receiver-imposed: AWGN, phase noise, IQ imbalance, DC offset, ADC. The same estimator run on the oracle capture defines the true observable phase and frequency; every error metric is computed against these. The tracking RMSE under this grading is 12.0–12.5 mrad, a factor ~ 100 above the self-referential figure. Distinct from the paper’s validation (Section 8, iv): here the closed-form claims are checked against an independent physical model rather than against their own assumptions.

10 Measured closed-loop error budget

Combining Sections 2–4, the steady-state closed-loop phase residual of the one-way loop is

$$\sigma^2 \approx \underbrace{\sigma_{\text{pn}}^2 f_s T}_{\text{white-FM walk per interval}} + \underbrace{(\sigma_{\omega} \ell T)^2}_{\text{frequency posterior} \times \text{latency}} + \underbrace{\sigma_{\text{track}}^2}_{\text{measurement/tracking}}. \quad (18)$$

Predicted terms at defaults: 45 mrad, 38 mrad (Section 4), and ~ 12 mrad. Ablation results (100 intervals):

Configuration	Tracking RMSE	Closed-loop residual RMS
Full defaults	12.0 mrad	69.6 mrad
RF impairments off (isolates the latency term)	1.1 mrad	34.3 mrad
Zero latency (isolates the tracking term)	12.3 mrad	12.3 mrad

The impairment-free case measures the latency term alone at 34.3 mrad against the 38 mrad Riccati prediction; the impairment-free tracking RMSE of 1.1 mrad sits on the AWGN CRLB (5). The full-default residual exceeds the root-sum-of-squares of the three predicted terms (60 mrad) by the flicker and shadowing contributions not in (18). Consequences: above moderate SNR the floor is set by oscillator quality, correction cadence, and latency, not pilot energy; the levers are a quieter reference, a shorter interval, or lower latency. Relative to (16): the latency term (12) has no counterpart there (updates are instantaneous); the Kalman steady-state posterior replaces per-measurement CRLBs; flicker and process continuity are added. A literature pass over 19 primary sources (each claim verified by three-referee adversarial review against full text) found no published closed-loop residual containing the latency-coupled term through December 2025; the closest works are Eq. (16) itself and open-loop drift-versus-update-interval analyses (Mghabghab et al., IEEE Access 2021).

11 Two-way synchronization and the π ambiguity

Implementing (9) in the simulator: forward and reverse frames traverse the same channel-tap realization and the same shadowing state within the interval (reciprocity), with the intra-frame LO walk of the forward frame folded into the oscillator state before the reverse frame so the two directions see one continuous noise process. The EKF consumes the half-difference phase and half-difference frequency (channel Doppler is common to both directions and cancels from the frequency too), with half the one-way measurement covariance. Result: the raw oscillator residual — unresolvable at -2.92 rad under one-way sync — becomes **81.8 mrad RMS** (two-station gain 99.83%) at 19.1% airtime.

The modulo- π branch of the half-difference is chosen at acquisition (candidate nearest zero) and thereafter the candidate nearest the filter prediction. A wrong acquisition branch produces a lock at relative phase π in which every subsequent measurement is self-consistent: the filter tracks its shifted belief exactly, and nothing internal to the loop reveals the error. This project first encountered the failure only after many correct runs, for an arithmetic reason worth recording: the default CFO times the interval, $1500 \times 0.05 = 75.000$ carrier cycles, is an exact integer, so the initial 1.2 rad offset ($< \pi/2$) reappeared unchanged at every acquisition and the nearest-zero rule happened to be right. A later configuration with 5 ms sub-intervals produced 7.5 cycles per step, the acquisition-time phase landed at 1.94 rad ($> \pi/2$), and the loop locked at π with 0.01% coherent gain. The deployed resolution is a one-time calibration after lock: any coarse external power measurement tells constructive from destructive (one bit); if destructive, the NCO is flipped by π . Implementation detail with a physical meaning: the filter must *not* be reset at the flip. The half-difference measurement is invariant under π shifts of the true phase, so the filter’s near-zero state simply re-attaches to the true branch; resetting it by π re-creates the shifted self-consistent lock (this was observed before being understood).

12 Reimplementation of DFPC/KF-DFPC and the comparison

The paper’s algorithms were implemented at two levels. *Statistics level*, faithful to Algorithms 1–2 including the Metropolis–Hastings weights, the graph sampler, and the Eq. 39 covariance mixing: at their canonical

setup ($N=20$, connectivity 0.2, $T=0.1$ ms, 0 dB) DFPC converges to 8.0° against the (16) bound of 5.9° (consistent: moderate connectivity sits near, sparse below the bound is not expected), and KF-DFPC reduces it to 2.1° — both of the paper’s qualitative claims reproduce. *Physical level*, at $N=2$ over the full simulator: the two-node Metropolis–Hastings weight is $w_{12} = \frac{1}{2}$, so each node retunes its own NCO by half its measured offset toward the other, with the same latency, quantization, and calibration machinery as the project’s own loops.

12.1 Anti-phase capture of naive over-the-air consensus

Run as published — each node consensus-averages its *raw one-way* measurement, consistent with the paper’s assumption that channel phase is handled elsewhere — the update in the relative coordinate $r = \theta_A - \theta_B$ is

$$r' = r + \frac{1}{2} \text{wrap}(-r + \phi_c) - \frac{1}{2} \text{wrap}(r + \phi_c), \quad (19)$$

since node A adds half its measurement $\text{wrap}(-r + \phi_c)$ to itself and node B adds half of $\text{wrap}(r + \phi_c)$. Fixed-point analysis:

- If neither argument wraps, the wraps are identities and $r' = r - r = 0$: the reciprocal channel phase enters both nodes’ updates identically and cancels; the pair aligns in one step regardless of ϕ_c .
- $r = \pi$ is also a fixed point: $\text{wrap}(-\pi + \phi_c) = \text{wrap}(\pi + \phi_c)$ because $-\pi \equiv \pi$, so the two halves cancel and $r' = \pi$. Perturbations decay back (the linearization about π matches that about 0), so anti-phase is *stable*.
- If exactly one argument crosses $\pm\pi$ during convergence — which happens when $|r| + |\phi_c| > \pi$ — the two wraps disagree by 2π and the update lands on the π branch.

Worked example at the default realization ($\phi_c = +2.922$, $r_0 = 1.2$): $\text{wrap}(1.2 + 2.92) = -2.16$, $\text{wrap}(-1.2 + 2.92) = +1.72$, $r_1 = 1.2 + \frac{1.72}{2} + \frac{2.16}{2} = 3.1416 = \pi$. Verification was threefold: the full simulation reproduces $r_1 = -3.117$ (the same point on the circle, 25 mrad of measurement noise from π), byte-identically across reruns; a four-line iteration of (19) with no radio code shows capture for $\phi_c = \pm 2.92$ and alignment for $\phi_c = -1.0, +0.5$; and across seeds, capture occurs exactly when $|r_0| + |\phi_c| > \pi$ predicts (channel phase +2.92: capture; $-1.50, +1.37, -1.39$: alignment). The failure is invisible to the paper’s channel-free model and is realization-dependent: the same hardware and algorithm aligns on one channel and locks anti-phase on another.

12.2 Head-to-head results, identical physical conditions

Approach	Residual RMS	2-station gain	Airtime
Two-tier micro-pilot (§13)	28.1 mrad	99.98%	26.0%
Hybrid one-way/anchors (§14)	33.8 mrad	99.97%	14.9%
Two-way EKF (§11)	80.8 mrad	99.84%	19.1%
KF-DFPC + reciprocity (paper, steelman)	82.8 mrad	99.83%	19.1%
DFPC + reciprocity (paper, unfiltered)	153.0 mrad	99.42%	19.1%
DFPC naive (paper, as published)	3009 mrad	0.68%	19.1%

The “+ reciprocity” rows exchange measurements over the side channel the paper already assumes for KF-DFPC and consense on the channel-free half-difference. Three findings, each independently checked:

1. Unfiltered consensus passes raw frequency measurement noise straight into the NCOs each interval. The measured steady frequency residual is 0.442 Hz; the implied drift per 50 ms interval is $2\pi \times 0.442 \times 0.05 = 139$ mrad, which with the 45 mrad walk predicts $\sqrt{139^2 + 45^2} = 146$ mrad against 153 observed. The paper’s claim that filtering matters at short intervals reproduces over a physical link.

2. Filtered consensus and the project’s two-way EKF land on the same floor (82.8 vs. 80.8 mrad; within a few mrad at every tested seed; not bit-identical, ruling out shared state). Both are Kalman-filtered reciprocal loops differing only in where the correction is applied. At fixed physics, the algorithm layer is interchangeable; the floor is (18).
3. The naive variant’s collapse (0.68% of ideal gain) is the measured cost of the channel-free measurement assumption.

13 Two-tier micro-pilot synchronization

Both dominant terms of (18) scale with the correction period, and phase — unlike detection, timing, or wide-range CFO — does not need a long pilot once the loop is locked: a short known sequence, correlated at the expected arrival after derotating by the current frequency estimate, gives a phase measurement with variance $1/(2\text{SNR}L_\mu)$ plus the (much smaller) intra-pilot phase noise. The two-tier scheme keeps the full two-way frame once per 50 ms interval (detection, timing, CFO, and the anchor for acquisition) and inserts M reciprocal phase-only micro-pilots — a 32-sample cyclic prefix plus a ZC-255, 287 samples = 0.29 ms per direction — at evenly spaced sub-intervals, with corrections issued every sub-interval. Tracking-mode feasibility at the defaults: timing drift per 10 ms sub-interval is 0.016 samples (search ± 4 suffices); derotation error from a 0.2 Hz residual CFO over 287 μ s is 0.36 mrad. Measured residual–airtime trade:

Micro-pilots per interval M	0	2	4	9
Residual RMS (mrad)	71.5	35.3	27.9	22.8
Airtime	19.1%	22.6%	26.0%	34.6%

The scaling follows (18) with $T \rightarrow T/(M+1)$: at $M = 4$ the predicted walk term is $2 \times 10^{-4} \sqrt{10^4} = 20$ mrad and the latency term $0.77 \times 0.01 = 8$ mrad, so $\sqrt{20^2 + 8^2 + \text{track}^2} \approx 23\text{--}25$ mrad against 27.9 measured (the full-frame substep and flicker account for the difference). Returns diminish beyond $M \approx 4$ as the measurement and staleness terms take over. It was the $M = 9$ configuration (5 ms substeps, 7.5 carrier cycles per step) that exposed the acquisition-branch failure of Section 11.

14 Hybrid one-way/two-way calibration

Reciprocal exchange costs both directions’ airtime, and (9) shows what the second direction buys: separation of θ from ϕ_c . For fixed stations, ϕ_c is nearly constant (the TDL taps are static at zero speed; shadowing changes amplitude, not phase) while θ wanders at the oscillator rate. The processes have disjoint spectral support, so a joint estimator can track both with mostly one-way data. State $x = [\theta, \omega, \phi_c]^\top$; dynamics diagonal except the θ – ω integrator; process noise q_θ (walk + oscillator), q_ω (random walk + flicker innovation), and $q_c = \sigma_c^2$, the assumed channel-phase drift per interval — the estimator’s statement of channel coherence time.

Observations:

- **One-way pilots** (a full frame each interval for timing/CFO; one-way micro-pilots at sub-intervals): phase observation $\theta + \phi_c$, Jacobian row $[-\sin(\cdot), 0, -\sin(\cdot)]$ — identical sensitivity to θ and ϕ_c . The direction $\theta - \phi_c$ is unobservable from one-way data alone; between anchors, the split is maintained only by the process priors (innovations are attributed in proportion to q_θ vs. q_c).
- **Two-way anchors** every K intervals: the half-difference observes θ (branch-resolved against the filter), the half-difference of directional frequencies observes ω free of channel Doppler, and ϕ_c is recovered as $m_f - \hat{\theta}$, which algebraically equals the half-sum (9) and is uncorrelated with the half-difference.

NCO corrections act on $[\theta, \omega]$ only (the channel cannot be actuated). At a π -calibration flip, the true θ moves by π while the one-way observable $\theta + \phi_c$ must remain matched to the filter’s; the implementation shifts the channel state by $-\pi$, keeping the tracked sum consistent.

14.1 Validation on both sides of the claim

Static channel. Residual versus anchor spacing K : 31.8 mrad at $K=1$ (22.6% airtime), 32.9 at $K=5$ (14.9%), 34.0 at $K=20$ (13.5%); the fully two-way control is 27.9 mrad at 26.0%. A $20\times$ change in reciprocity cadence moves the residual by 2 mrad: anchor cadence has decoupled from oscillator quality.

Moving channel. Doppler at 915 MHz is $f_D = 3.05$ Hz per m/s; TDL-D’s dominant line-of-sight tap acquires a quasi-constant Doppler within $\pm f_D$, so ϕ_c drifts up to $2\pi f_D T$ per interval — 0.19 rad at 0.2 m/s, 0.48 rad at 0.5 m/s, against the static prior $\sigma_c = 0.01$. Sweep (60 intervals, seed 0):

speed (m/s)	f_D (Hz)	scheme (K , channel prior)	residual	gain
0.2	0.61	control (fully two-way)	43 mrad	99.95%
0.2	0.61	hybrid $K=1$, static prior	68 mrad	99.88%
0.2	0.61	hybrid $K=5$, static prior	446 mrad	95.3%
0.2	0.61	hybrid $K=5$, matched prior	235 mrad	98.7%
0.2	0.61	hybrid $K=20$, static prior	1752 mrad	55.1%
0.5	1.52	control (fully two-way)	33 mrad	99.97%
0.5	1.52	hybrid $K=1$, static prior	128 mrad	99.6%
0.5	1.52	hybrid $K=1$, matched prior	44 mrad	99.95%
0.5	1.52	hybrid $K=5$, matched prior	208 mrad	98.9%
0.5	1.52	hybrid $K=20$, matched prior	530 mrad	93.4%

Readings. The control is immune: its measurements are channel-free at every substep. Under the static prior, the filter attributes the channel drift to the oscillator (larger q_θ wins the innovation) and chases the channel with NCO corrections; degradation is monotone in both Doppler and K . The matched prior $\sigma_c \approx 2\pi f_D T$ recovers fully at $K=1$ (44 vs. 33 mrad control) and partially at sparse K : correctly so, because the prior changes attribution but cannot supply the separating information only anchors carry. Together the two experiments establish the design rule: **reciprocity cadence is set by channel coherence time, not oscillator quality** — on static links, anchors at 1 Hz or slower suffice and total sync airtime falls to 13–15%; at walking-speed Doppler, anchors are needed at the interval rate. A known residual defect: the one-way frequency observation measures ω plus the channel Doppler and is therefore biased at nonzero speed; a fourth state (channel Doppler rate) would absorb it and is the next refinement.

15 Application split: CSI-aided versus open-loop coherence

The identifiability analysis of Section 3 divides applications of these loops into two classes, and the simulator quantifies both.

CSI-aided (closed-loop) joint transmission. When a user feeds back channel estimates, the two-station combining gain at the user depends only on the *differential* station phase relative to its value at the last CSI epoch: all static biases, including the one-way channel bias, cancel. Evaluating the one-way loop’s closed-loop oscillator residual as the differential-phase process, with pilot estimation noise added per epoch, gives mean JT gain versus CSI refresh cadence: 100.00% (refresh every interval), 99.88% (2), 99.79% (5), 99.72% (10), 99.56% (20 intervals = 1 s). One-way synchronization plus ordinary CSI is sufficient for cell-free-style coherent service, with CSI staleness to 1 s costing under 0.5% of gain.

Open-loop coherence. For passive coherent sensing there is no feedback path; the carriers must align at the antennas, which requires the two-way or hybrid loops. Their bounded residual has a consequence specific to sensing: a *closed-loop* bounded phase error (28–82 mrad, never diverging) costs a fixed 0.1–0.7% of coherent gain regardless of integration time, whereas open-loop (uncorrected) oscillators decorrelate and cap the usable dwell. Integration time for weak-target detection becomes limited by target and channel dynamics, not by the stations’ clocks.

16 Summary and open items

1. Grading discipline: estimator errors must be computed against an independent ground truth (the oracle path), not against the estimator’s inputs; the difference was a factor of ~ 100 here.
2. The closed-loop floor (18) is predicted by two hand-computable terms (walk $\sqrt{kT}$, latency $\sigma_{\hat{\omega}}\ell T$) and verified by ablation to within a few mrad. The latency term has no counterpart in the published literature through Dec. 2025 (19-source verified search).
3. The reference paper’s algorithms reproduce at statistics level (residuals vs. its Eq. 27) and, given reciprocity, over a physical link, where filtered consensus and the project’s EKF loop coincide at the physical floor (82.8 vs. 80.8 mrad). Applied as published, the wrapped symmetric consensus is bistable over real channels (19) and locks anti-phase on realizations satisfying $|r_0| + |\phi_c| > \pi$; verified by hand arithmetic, by a radio-free iteration, and across seeds.
4. Cadence, not energy, moves the floor: two-tier micro-pilots reach 22.8–28 mrad ($3\times$ the plain loop) for 7–15% added airtime; hybrid one-way/two-way calibration holds 33 mrad at 13–15% airtime on static channels, with reciprocity cadence set by channel coherence (validated under Doppler in both directions, including the failure of a mismatched channel prior).
5. Open: a fourth state for channel Doppler; $N > 2$ consensus over physical links with wrap-safe (phasor-domain) updates; a delayed-feedback PLL literature check before claiming the latency term in print; hardware validation — every number above is simulation.

Sources

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- Repository documents: LITERATURE_REVIEW.md (verified prior-art pass and novelty assessment); paper_review_rashid_simulation_design.pdf; README.md; hybrid_calibration/README.md. All numbers in this document regenerate from simulation.py (models sdr, twoway, micro, hybrid, dfpc, kdfpc, compare) and pytest (18 tests).